Joint Fusion and Conditional Intervention Value Learning for Cooperative Multi-USV Target Defense

Manuscript in preparation

***Abstract*—Cooperative target defense requires the control choices of individual vessels to form a successful team maneuver. We address two distinct decisions in learned–Boids controller fusion: which combination of fusion coefficients the fleet should execute, and when changing a local fusion decision improves the subsequent defense outcome. First, a candidate-interaction model predicts the motion induced by each vessel's admissible actuator mixtures and selects a joint configuration using their pairwise consequences. Second, a conditional intervention value model learns a shared task potential from complete, paired feedback continuations. Single-vessel differences identify useful changes within a joint configuration; separate rescue and harm predictions, factual calibration, and intervention timing guide deployment. With a fully trained six-vessel source, the complete method achieves 97.0% success on 856 development episodes for training seed 42, exceeding ARBoids by 29.4 points and the residual policy by 57.1 points. Matched comparisons separate an 8.1-point joint-selection gain, a further 4.1-point intervention gain, and a 3.0-point advantage over an ordinary-return MLP with a development-selected decision threshold. On new states, conditional value ranks randomly sampled interventions correctly in 73.5% of informative comparisons, versus 49.2% for the MLP. These results distinguish joint-selection gains, intervention gains, and value-ranking accuracy; complete independent training and held-out evaluation determine their final empirical scope.**

***Index Terms*—Multi-USV target defense, cooperative control, controller fusion, joint decision making, conditional intervention value.**

I. Introduction

A fleet defending a target against a faster intruder must coordinate how its vessels move together. Individually plausible maneuvers can send two defenders through the same narrow space, leave an interception direction uncovered, or disturb an otherwise successful approach. The central decision is therefore how available controls should be combined across the fleet, and which changes are worth making as the encounter unfolds. Target guarding and cooperative pursuit provide the task setting [2], [3], [5]; learned marine control provides flexible individual maneuvers [6], [10], [11].

ARBoids [18] blends a learned SAC controller with a Boids controller using an adaptive coefficient. This preserves a structured coordination prior while allowing task experience to change the maneuver. We retain these two control sources and their original physical execution model. Our first contribution concerns the *joint choice of mixtures*: a vessel's preferred coefficient depends on which control candidates its teammates execute. Observing teammate positions alone does not resolve this dependence. We predict the motion associated with concurrent candidate controls and select a combination using the resulting interaction costs.

A joint selection criterion still does not determine whether a local change helps the final task. Avoiding an imminent encounter may delay capture; an early change may help where the same change after the intruder passes the fleet cannot. Our second contribution concerns the *conditional value of changing a fusion decision*. We compare complete feedback continuations that differ in a specified vessel's mixture while retaining the other components of the current joint plan. Both branches encounter the same sampled future disturbances and then follow the same frozen controller on their own evolved states. These comparisons train a common task potential, so a local value difference remains consistent with the value of a complete fleet intervention.

The resulting method connects physical interaction with task consequence. Joint fusion coordinates the current candidate controls. Conditional value learning decides whether to keep that configuration or execute a finite intervention, including a delayed intervention when the opportunity is more favorable. The two mechanisms have separate experimental questions: joint selection is compared with independent selection under the same candidate predictions, and conditional value learning is compared with ordinary absolute-return learning under the same observation, parameter, update, and deployment budgets.

The contributions are:

- A joint fusion rule that resolves interactions between actuator-level learned–Boids candidates while preserving the original control hull.
- A conditional intervention value model that connects paired single-vessel consequences to a shared complete-plan task potential, with a measured factual reference and explicit intervention timing.

We test these contributions through separate comparisons of whole-method performance, matched joint and independent selection, and conditional versus ordinary return learning. Paired successes and failures identify the rescues and harms behind each increment.

II. Task and Control Model

A. Dynamics and task outcomes

The numerical environment uses the released ARBoids vessel dynamics and task implementation [18]. A defender obeys

$$\dot{\eta}_i = R(\psi_i)\nu_i, \qquad M\dot{\nu}_i + C(\nu_i)\nu_i + D(\nu_i)\nu_i = \tau_i. \quad (1)$$

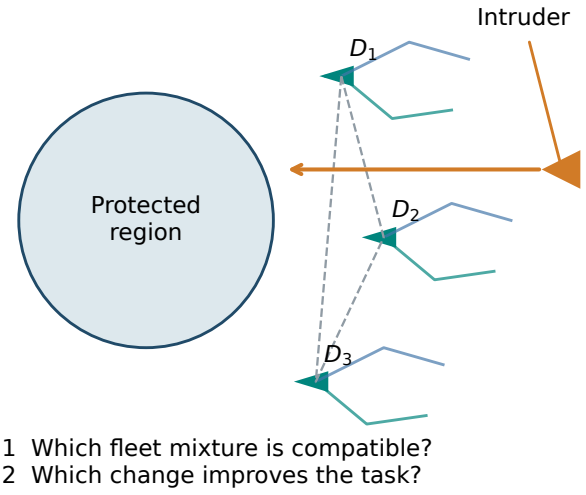


Fig. 1. Target defense and its two decision questions. The fleet must choose mutually compatible mixtures and determine whether a local change improves the eventual outcome. The drawing is schematic; measured trajectories appear in the mechanism and VRX comparisons.

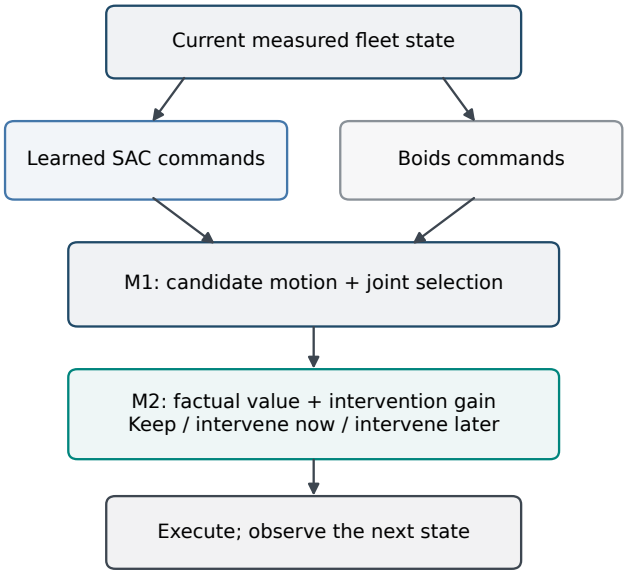


Fig. 2. Execution path. The original learned and Boids controllers produce current commands. Mechanism 1 jointly selects their mixtures. Mechanism 2 evaluates finite interventions against that factual controller; accepted interventions retain the same feedback control sources throughout their execution window.

The integration period is 0.05 s and the command period is 0.2 s. Each defender thruster remains in $[-500, 1000]$ N; the agility parameter scales the intruder's thrust bounds. The protected radius is 15 m, and both capture and defender-collision thresholds are 5 m.

We preserve the source termination order. Intrusion failure occurs when the intruder enters the protected radius *or* its distance to the target becomes smaller than that of every defender. Defender collision is checked next, followed by capture, and then successful defense at the 80-s deadline. The radial-pass condition is therefore included in reported source failures, even when the intruder has not physically crossed the target boundary. We additionally record physical boundary crossings, captures, collisions, and episode duration.

The controller observes current vessel geometry, measured velocity, and the available learned and Boids commands. It does not observe future disturbances or the intruder's future control sequence. Source dynamics, source policies, and the source opponent are shared across matched comparisons. The source APF opponent has an order-dependent obstacle update; canonical feature ordering does not change that simulator behavior.

B. Control sources and admissible mixtures

Let u_i^{L} and u_i^{B} be the current learned and Boids thrust vectors. We use two actuator coefficients,

$$u_i = g_i \odot u_i^{\mathrm{L}} + (1 - g_i) \odot u_i^{\mathrm{B}}, \qquad g_i = (g_{i,1}, g_{i,2}) \in [0,1]^2. \tag{2}$$

Allowing separate port and starboard coefficients gives the same admissible representation to joint and independent selectors. Both coefficients remain within the original componentwise control hull. The source adapter supplies a scalar preference, which is mapped to a bounded center $\bar{g}_i$ and shared by the two actuators. Each actuator chooses from $\{0, \bar{g}_i, 1\}$, giving nine candidates per vessel. Duplicate candidates are coalesced.

The primary comparisons train the source policies directly with six vessels for one million steps. Joint selection, independent selection, and both value learners reuse the same selected ARBoids source weights. The source policy's mean teammate embedding permits observations from other fleet sizes. Earlier experiments transferring a three-vessel source to six vessels are reported as a separate supporting comparison.

III. Mechanism 1: Joint Selection of Fusion Candidates

A. Candidate motion and pairwise interaction

For every candidate g_i^k, a public-state nominal model predicts the vessel trajectory over 3.2 s at 0.1-s prediction steps. Learned and Boids commands are recomputed every 0.2 s on predicted observations. The candidate coefficient is retained over this forecast, and the executed fleet replans at the physical command rate. This predictor describes the motion associated with candidate intent; it does not replace the complete task continuations used for value learning.

For two candidates, define

$$d_{ij}^{k\ell} = \min_{\tau \in [0,H]} \|\hat{p}_i^k(\tau) - \hat{p}_j^\ell(\tau)\|, \qquad \psi_{ij}(k,\ell) = [7 - d_{ij}^{k\ell}]_+^2. \tag{3}$$

The unary term penalizes departure from the source preference,

$$\phi_i(k) = \frac{1}{2} \sum_{a=1}^{2} \left(\frac{g_{i,a}^k - \bar{g}_i}{0.4} \right)^2. \tag{4}$$

The joint fleet configuration minimizes

$$\boldsymbol{k}^* \approx \arg\min_{\boldsymbol{k}} \left[\sum_i \phi_i(k_i) + 32 \sum_{i<j} \psi_{ij}(k_i, k_j) \right]. \quad (5)$$

The interaction term depends on both vessels' selected candidates. Its role is to distinguish complete combinations that can have similar individual preferences but different physical compatibility.

B. Joint search and the matched independent selector

A width-128 beam constructs complete configurations, followed by three coordinate sweeps. The refined selector additionally refines every retained beam endpoint and performs one pair-block sweep. It accepts a refinement only when both total preference departure and total predicted interaction cost are no worse than the original joint solution. This improves search within the same candidate library and objective.

The independent comparator receives the identical candidates, trajectories, and pairwise cost tensor. With source-preference probabilities $p_j(\ell) \propto \exp[-\phi_j(\ell)]$, it chooses

$$k_i^{\mathrm{ind}} = \arg\min_{k} \left[\phi_i(k) + 32 \sum_{j \neq i} \sum_{\ell} p_j(\ell) \psi_{ij}(k, \ell) \right]. \quad (6)$$

Thus both methods account for the same predicted candidate interactions. The comparison isolates the consequence of selecting a joint configuration instead of minimizing expected cost independently. Joint optimization has additional search cost, which is reported separately from prediction cost.

IV. Mechanism 2: Conditional Intervention Value

A. A complete intervention and its factual continuation

Let π_0 be the frozen predecessor: either the source fusion controller or the joint fusion controller. An intervention $z = (\boldsymbol{c}, d, h)$ specifies actuator mixtures, delay d, and duration h, measured in physical control ticks. For each vessel, code zero retains the predecessor's decision; the other nine codes specify actuator pairs in $\{0, 0.5, 1\}^2$. Before the delay and after the intervention window, the controller follows π_0. During the window, its learned and Boids commands still update from the current state while the specified mixture is held. Unchanged vessels continue to respond through the predecessor.

For public context x and sampled future ω, let

$$Y(x, z, \omega) = \mathbb{I}\{\text{complete continuation succeeds}\}, \quad (7)$$

$$V(x, z) = \mathbb{E}_\omega[Y(x, z, \omega)]. \quad (8)$$

The factual option z_0 follows π_0 throughout. Two complete options z, z' define a single-vessel comparison when they share timing and all vessel codes except one. Their measured conditional difference is

$$\widehat{\Delta}(x; z, z') = \frac{\sum_r w_r [Y(x, z', \omega_r) - Y(x, z, \omega_r)]}{\sum_r w_r}. \quad (9)$$

Common futures pair the outcomes; later controls are recomputed on each branch's evolved state. Full terminal outcomes include failed as well as successful branches. Sampling may emphasize development failures, but deployed decisions receive only the current context.

B. One task potential for local differences and complete plans

A candidate-conditioned graph network represents a shared scalar potential $V_\theta(x, z) \in [0, 1]$. Node features contain vessel state, both source controls, the factual control, the intervention code, its physical command change, and observable vessel identity. Nominal motion descriptors at 1, 2, and 4 s encode each candidate's position, velocity, heading, and change from factual motion. Pair messages combine relative geometry and predicted separation with both vessels' candidate-conditioned representations. Fleet pooling and timing features produce the complete-plan value. Local differences use this same potential,

$$\Delta_\theta(x; z, z') = V_\theta(x, z') - V_\theta(x, z). \quad (10)$$

Consequently, reversing a comparison reverses its sign, and differences sum to zero around a closed comparison cycle. The method does not add independent local scores to approximate a fleet outcome.

The learning objective combines complete outcomes with their conditional differences:

$$\begin{aligned} \mathcal{L} = \mathcal{L}_Y + 4\mathcal{L}_{\mathrm{local}} + 2\mathcal{L}_{\mathrm{factual}\Delta} + \mathcal{L}_{\mathrm{rank}} \\ + 4\mathcal{L}_{\mathrm{factual}} + \mathcal{L}_{\mathrm{bank}} + \tfrac{1}{2}\mathcal{L}_{\mathrm{stratum}}. \end{aligned} \quad (11)$$

Here $\mathcal{L}_Y$ is binary outcome cross entropy, $\mathcal{L}_{\mathrm{local}}$ fits same-future single-vessel differences, and $\mathcal{L}_{\mathrm{factual}\Delta}$ fits differences from the factual option. The ranking term uses outcome-discordant pairs while retaining neutral pairs in its normalization. Factual and bank losses anchor observed factual outcomes and additional states from complete factual trajectories. The stratum loss fits intervention outcomes conditional on factual success or failure. Sampling corrections retain the chosen comparison population.

Three width-64 models receive 10,000 updates each, for 30,000 total updates and 146,319 fitted parameters including six factual-calibration parameters. The matched ordinary-return comparator is one three-hidden-layer ReLU MLP with width 233 and 146,092 parameters. It receives the same measured outcomes, factual-state observations, candidate catalog, and 30,000 updates. Its objective is pooled absolute-outcome cross entropy. Its deployment threshold is selected on the common development screen. The graph representation, paired objectives, factual calibration, and local support model are constituents of the proposed mechanism; matching network parameter count does not equate their extra calibration storage or inference cost.

The conditional potential separates rescue from harm. With factual success probability $p_0(x)$, it estimates $r_\theta(x, z) = P(Y_z = 1 \mid Y_0 = 0, x)$ and $h_\theta(x, z) = P(Y_z = 0 \mid Y_0 = 1, x)$ from shared-future outcomes. The complete potential and its factual difference become

$$V_\theta(x, z) = (1 - p_0) r_\theta + p_0 (1 - h_\theta), \quad (12)$$

$$\Delta_\theta(x; z_0, z) = (1 - p_0) r_\theta - p_0 h_\theta. \quad (13)$$

This decomposition distinguishes a possible rescue from damage to an already successful continuation. Three layout folds keep all paired futures and factual trajectory states of a layout together. Each ensemble member excludes one fold from gradient fitting and uses its factual observations to calibrate

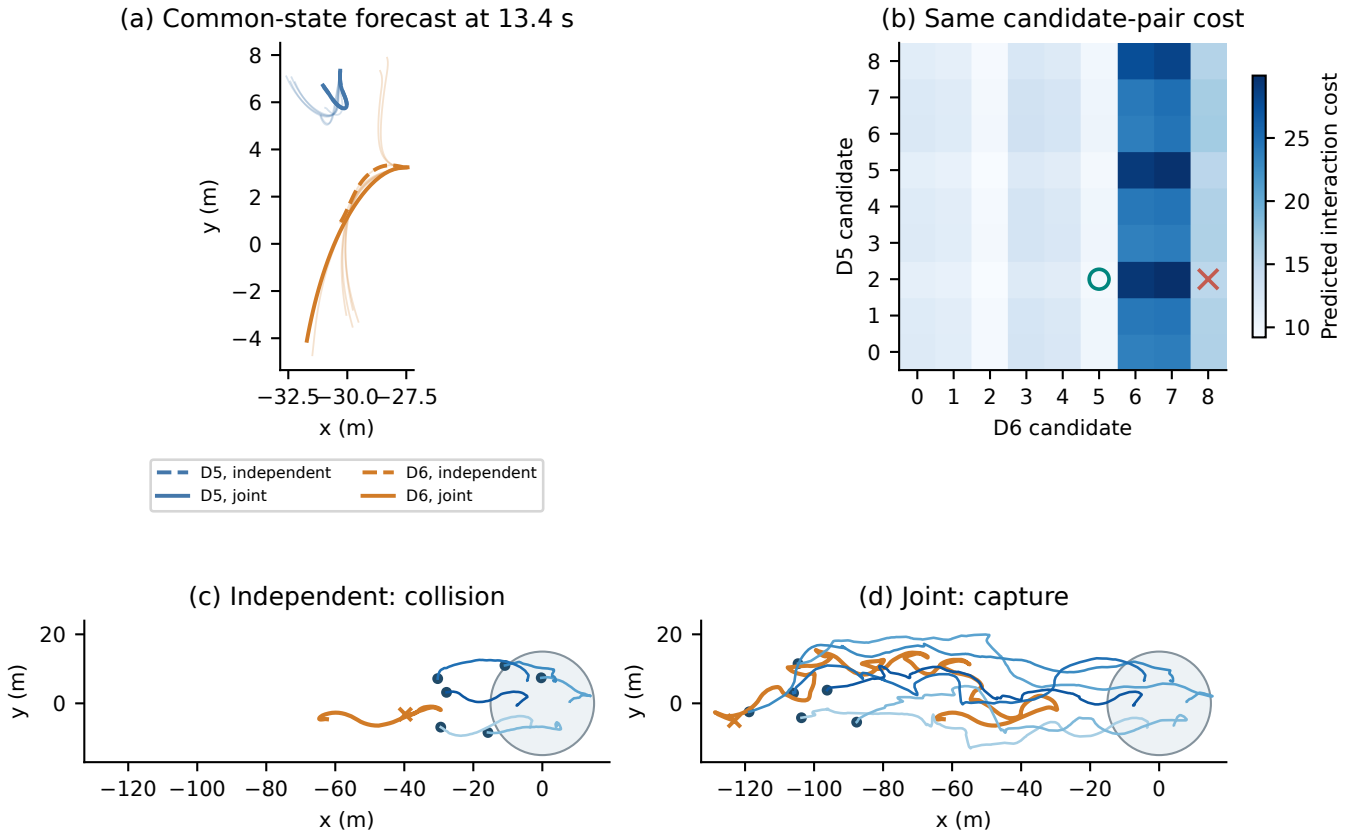


Fig. 3. Mechanism 1 with the fully trained six-vessel source on development scene 431020053. The top panels compare identical candidate forecasts at a common state at 13.4 s on the independent trajectory, where the selected pair has its largest predicted risk reduction. The complete joint objective decreases from 495.5 to 339.0. The bottom panels replay both policies from the same initial state and future: independent selection collides at 13.6 s, whereas joint selection captures at 70.6 s. This illustration is the first joint rescue of an independent collision in the fixed 4310 row order.

the factual probability through a bounded monotone map. The refit and its fresh ordinary MLP share the new predecessor's measured training data, candidate catalog, and 30,000 updates; their parameter counts are 146,319 and 146,092, respectively. For the six-vessel source, 743 paired roots and 856 factual trajectories supply 36,900 unique terminal observations and 58,797 supervised labels. Their collection uses 4,353,021 physical simulation steps. The ordinary controller is also evaluated with positive gain margins of 0.02, 0.08, and 0.12; its reported tuned result uses the best decision rule on the common 228-episode development screen without additional network updates.

C. Factual calibration, timing, and execution

A measured factual bank estimates the success probability of retaining π_0 in a nearby context. Together with layout-disjoint neural calibration, this estimate sets the mixture weights of rescue and harm in the complete task potential. Locally measured paired residuals then correct its value differences. Rescue and harm observations, ensemble disagreement, and distance from observed support contribute to the decision penalty.

An intervention that starts after factual termination cannot change that outcome. For delay d, factual trajectory data provide estimates of survival until d, separately for factual success and failure. These estimates bound possible rescue and harm and prevent a delayed option from claiming an opportunity that has already ended. The deployed rule is

$$z^* = \arg\max_{z \neq z_0} \{\widehat{\Delta}(x; z_0, z) - \beta U(x, z)\}, \tag{14}$$

and executes z^* only if this score exceeds a positive margin. Otherwise it retains the factual controller and revisits the decision on the next eligible observation. The reference implementation queries every five physical ticks and accepts at most one intervention per episode. The ordinary MLP shares this observation clock and intervention budget.

The uncertainty penalty is a decision quantity, not a guaranteed coverage interval for an unseen state. Development importance weights specify objective mass rather than independent replications. New refits use effective future counts for the rescue/harm penalty. Replacing π_0 requires freshly generated states and terminal labels; outcomes measured under one predecessor are not relabeled as outcomes of another.

V. Experimental Design

The evidence follows three comparisons: the complete method against adequately trained relevant baselines; joint versus independent fusion with matched candidates and predictions; and conditional intervention value versus matched

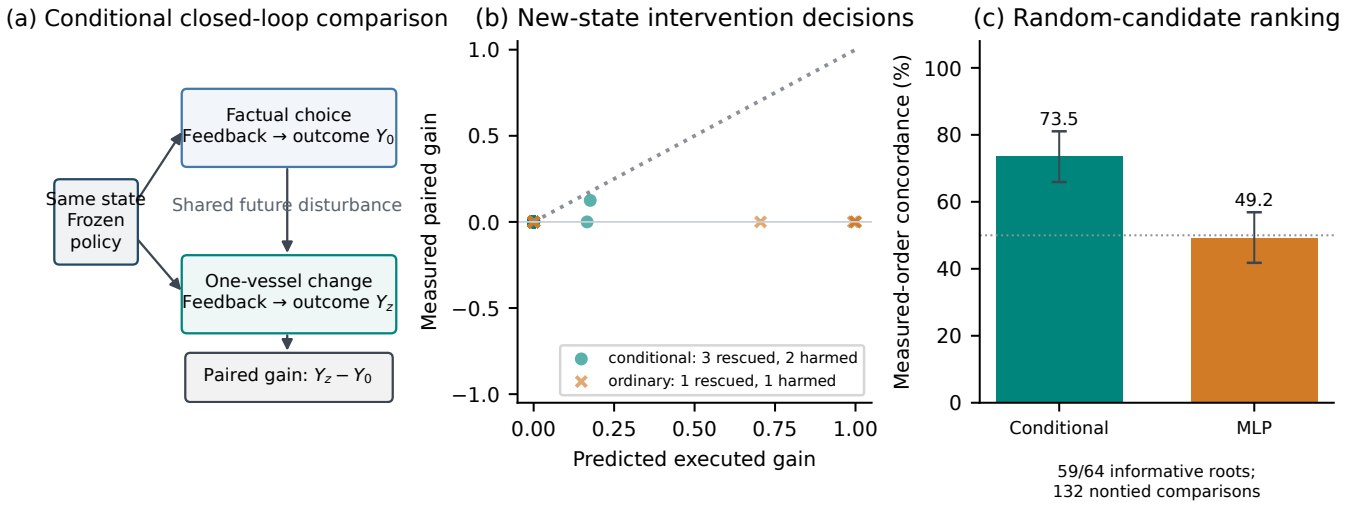


Fig. 4. Mechanism 2. Paired continuations share the initial state, other current decisions, frozen continuation policy, and sampled future. Their eventual outcomes supervise a shared task potential. Factual success, intervention delay, and measured support determine whether the selected change is executed. The decision and ranking panels use the six-vessel-source refit on 64 new states, with three outcome-independent random candidates and eight shared futures per state; error bars are 95% intervals from a stratified bootstrap over states.

ordinary return learning. Four configurations separate the two mechanisms: F00 uses neither mechanism, F10 uses joint fusion, F01 uses conditional value on the source predecessor, and F11 uses conditional value on the joint predecessor. Each value model is trained with continuations from its own frozen predecessor.

All primary contrasts pair the same initial layout, agility, fleet size, and future seed. Repeated futures from one layout remain in the same bootstrap cluster. Success differences, rescue/harm counts, collision, and capped duration are reported together. Capture-only duration is secondary because successful episodes can differ across methods. Independent learning repetitions are distinguished from additional evaluation seeds and additional future samples.

The current 856-episode development collection contains 256 episodes from the original source suite and 200 episodes in each of three later suites. The source suite repeats 64 layouts under four futures; the other suites contain new layouts. The 4310 suite did not enter the reference value-training data, but its repeated use in model selection makes it development evidence. The reserved 4322 layout family remains outside development and is evaluated only after the method and comparators are frozen.

Five independently initialized source-policy runs use seeds 42–46. Joint and independent fusion share the resulting checkpoint within each run. Each complete value-learning pipeline requires new factual and intervention continuations from its own frozen predecessor. Earlier value-only repetitions that share a source backbone are reported separately in Appendix A. Baseline SAC, residual policy, and ARBoids runs retain the source million-step training budget, with the fleet size and actual random seed recorded explicitly. Each completed baseline run supplies the checkpoint with the highest training-evaluation success rate, breaking ties by mean reward and retaining the first exact tie.

TABLE I

FULLY TRAINED SIX-VESSEL-SOURCE DEVELOPMENT COMPARISON ON THE SAME 856 PAIRED EPISODES, TRAINING SEED 42. F00–F11 USE THE COMMON MECHANISM EXECUTOR. EACH ORDINARY MLP SHARES ITS CONDITIONAL COUNTERPART'S PREDECESSOR, TRAINING OBSERVATIONS, CAPACITY, AND UPDATE BUDGET, AND USES A DEVELOPMENT-SELECTED GAIN THRESHOLD.

Method	Source /256	4310 /200	Earlier /200	Later /200	Total /856
SAC	42	27	20	29	118
RP	99	82	80	80	341
ARBoids	171	135	137	135	578
F00: source	171	135	137	136	579
F10: joint fusion	238	188	180	189	795
F01: conditional value	179	157	144	161	641
F11: complete	249	195	193	193	830
Independent fusion	220	167	166	173	726
Tuned MLP on F00	166	139	133	134	572
Tuned MLP on F10	246	190	183	185	804

VI. RESULTS

A. Complete method and the four configurations

The reference six-vessel-source method succeeds in 830 of 856 episodes (97.0%), compared with 795 (92.9%) for its own joint predecessor. It rescues 49 predecessor failures and harms 14 predecessor successes, adding 4.1 percentage points with a layout-cluster bootstrap interval of [2.5,5.8] points. The number of failed defenses falls from 61 to 26, a 57.4% relative reduction. Captures increase from 793 to 828, collisions decrease from nine to seven, and source-defined intrusion failures decrease from 52 to 19. On 779 episodes captured by both controllers, the mean capture-time difference is -0.51 s $[-0.86,-0.16]$. These development intervals describe resampling uncertainty and do not correct for adaptive model selection.

Without joint selection, conditional value raises F00 from 579 to 641 successes (67.6% to 74.9%). Its 103 rescues and

TABLE II
FIVE INDEPENDENT SIX-VESSEL SOURCE-TRAINING RUNS AND THEIR PAIRED FUSION SELECTORS. EACH CELL COUNTS SUCCESSFUL DEFENSES IN THE SAME 856 DEVELOPMENT EPISODES. JOINT AND INDEPENDENT FUSION REUSE THE ARBOIDS CHECKPOINT IN THEIR ROW.

Seed	SAC	RP	ARBoids	Independent	Joint
42	118	341	578	726	795
43	212	337	355	385	404
44	182	253	584	773	780
45	320	247	615	736	758
46	145	252	675	707	755

41 harms yield a separate 7.2-point increment [4.6,9.8]. The complete method also exceeds F01 by 189 successes, while F10 exceeds F00 by 216. Thus both mechanisms improve their corresponding predecessor in the same six-vessel-source four-configuration comparison. F00 uses the ARBoids checkpoint through the shared mechanism executor; its 579 successes are retained separately from the native baseline executor's 578. Appendix A preserves the earlier transferred-source comparisons and their adverse outcomes.

ARBoids trained from scratch with six vessels obtains 578/856 successes (67.5%) after the full million-step run. Its checkpoint is selected at 780,000 steps by training-evaluation success and reward. The complete six-vessel-source method adds 29.4 points [25.8,33.1], rescuing 261 baseline failures and harming nine baseline successes on paired development episodes. It also exceeds the fully trained residual policy by 57.1 points [53.3,60.8], with 496 rescues and seven harms. Relative to ARBoids, captures increase from 578 to 828 and collisions decrease from 270 to seven; source-defined intrusion failures increase from eight to 19. On the 568 episodes captured by both methods, the complete method takes 4.46 s longer [3.45,5.46]. This comparison establishes a substantial reliability gain with a capture-time tradeoff for this training seed.

All 15 baseline training runs complete one million steps and the full 856-episode endpoint comparison. Table II reports every training seed. ARBoids achieves $65.6 \pm 14.2\%$ success across seeds, RP $33.4 \pm 5.7\%$, and SAC $22.8 \pm 9.1\%$ (mean and sample standard deviation). The complete method's 97.0% result conditions on seed 42; it is not substituted for a five-run complete-method average.

B. Direct evidence for joint selection

With the fully trained six-vessel ARBoids source, joint selection raises standalone success to 795/856 (92.9%). Independent selection using the identical source weights, candidates, prediction resources, and refinement settings obtains 726/856 (84.8%). The joint gain is 8.1 points [5.3,11.1], comprising 112 rescues and 43 harms. Captures increase from 723 to 793, collisions decrease from 20 to nine, and source-defined failures from 110 to 52. At this predecessor's success rate, the largest possible additional success-rate gain is 7.1 points; the conditional-value refit realizes 4.1 of those points using its own freshly measured continuations. Across all five completed source-training runs, joint and independent success rates are $81.6 \pm 19.3\%$ and $77.7 \pm 18.5\%$, respectively (mean and sample standard deviation). Their per-seed joint-selection gains are 8.1, 2.2, 0.8, 2.6, and 5.6 points, averaging 3.9 ± 2.9 points. Joint selection improves every run, but its gain varies substantially with the learned source; Table II and Fig. 5(d) retain this variation.

On the same six-vessel source and 856 episodes, state-attention and candidate-attention replacements obtain 626 (73.1%) and 665 (77.7%) successes. Both use two attention layers of width 128, 192,769 parameters, 24,000 updates, and the same 24,576 measured teacher queries from teacher and student trajectories. The joint selector provides training supervision; each replacement makes its own neural choices at deployment. Conditioning each candidate on only the teammates' source-preferred reference trajectories yields 754 successes (88.1%), compared with 726 (84.8%) for independent expectation over teammate candidates and 795 (92.9%) for joint selection. Figure 6(a) therefore compares representations and decision dependencies under the same learned source. The transferred-source counterparts remain in Appendix A.

C. Conditional value versus ordinary return learning

Without joint selection, the conditional value model obtains 641/856 successes against 572/856 for its matched ordinary MLP, a gain of 8.1 points [4.9,11.2], with 118 rescues and 49 harms. The ordinary model uses a development-selected margin of 0.02; its unthresholded rule obtains 556 successes. Both estimators use the same 743 factual roots and paired continuations, approximately 146,000 parameters, and 30,000 updates. Collisions decrease from 277 to 209. This comparison establishes a direct conditional-value gain on the source predecessor, separately from its incremental contribution after joint selection.

An additional ordinary-return control uses the conditional model's candidate-interaction network, public motion features, candidate catalog, and exact training observations. Its 145,923 parameters match the conditional model's 146,319 within 0.3%; both receive 30,000 updates. It regresses absolute returns without paired-difference supervision or factual calibration. On a common batched executor, conditional value succeeds in 641/856 episodes, the graph-return control in 578/856, and the ordinary MLP in 571/856. The graph control's margin of 0.12 is selected on the common 228-episode development screen. Conditional value rescues 108 graph-control failures and harms 45 successes, gaining 7.4 points [4.6,10.3]; collisions decrease from 269 to 209. Thus the measured advantage also survives matching the candidate-interaction representation.

After joint selection, the conditional model reaches 830/856 versus 804/856 for the development-selected ordinary controller, an increment of 3.0 points [1.3,4.8]. The paired outcomes comprise 39 rescues and 13 harms. Captures increase from 801 to 828, collisions decrease from 15 to seven, and source-defined failures decrease from 37 to 19. On 788 jointly captured episodes, the paired capture-time difference is -0.48 s $[-0.91,-0.05]$. The ordinary model's unthresholded maximum-return rule yields 753 successes; the stronger 804-

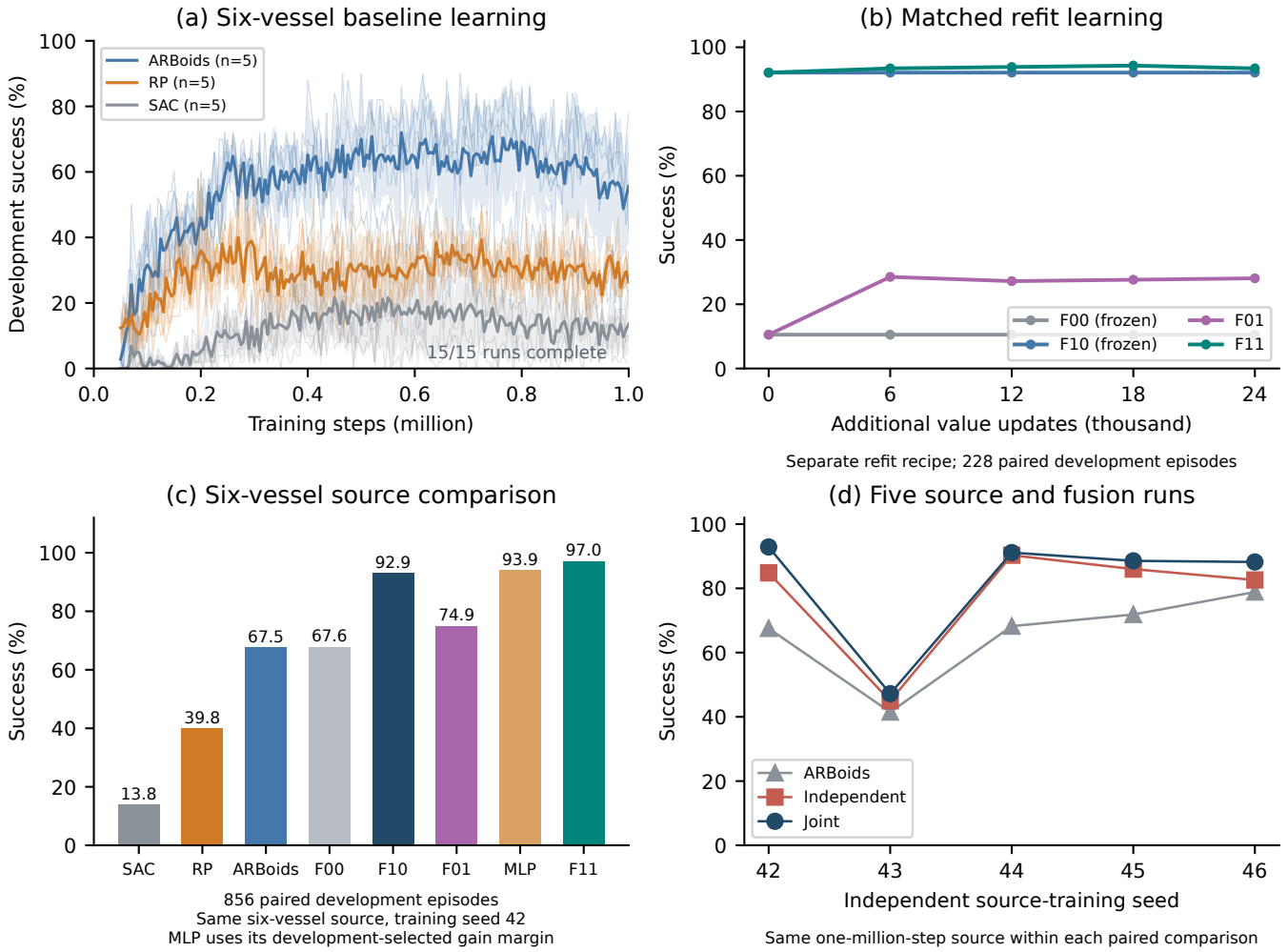


Fig. 5. Overall evidence. All 15 baseline learning curves complete one million steps. The refit-learning panel retains the separately identified transferred-source four-configuration experiment. The endpoint panel reports all four six-vessel-source configurations and the tuned ordinary MLP at training seed 42. The repetition panel compares ARBoids, independent fusion, and joint fusion across five independently trained source policies; the source checkpoint is shared within each seed.

success comparator, selected with margin 0.08, is used for the main mechanism claim.

For value validation, one eligible pre-action state is sampled uniformly from each of 64 complete factual trajectories on new layouts. Both learned choices, the factual action, and three outcome-independent random candidates are evaluated to termination under eight shared future seeds at every root. Among 132 nontied comparisons from 59 informative roots, conditional concordance is 73.5% [65.9,81.1], versus 49.2% [41.8,56.9] for ordinary return learning. The paired advantage is 24.2 points [14.9,33.6], and gain mean squared error is 0.068 versus 0.181. Executed conditional choices yield 459/512 successes, against 458 for both factual continuation and the tuned MLP. Conditional choices rescue three futures and harm two; ordinary choices rescue one and harm one. Thus the ranking improvement is substantial, while its realized gain at uniformly sampled decision times is one success. This state-level decision experiment and the 35 additional full-episode successes measure distinct deployment opportunities.

Applying the same new-state protocol under F00 gives 391/512 successful continuations for factual control, F01, and its tuned ordinary MLP. Conditional decisions rescue 35 futures and harm 35; the ordinary model leaves all outcomes unchanged. F01's random-candidate concordance is 64.0%, compared with 54.7% for ordinary value, with a paired difference of 9.4 points [−1.7,19.1]. The full-episode improvement therefore coexists with unresolved timing and harm control at uniformly sampled states.

The transferred-source component comparison isolates intervention timing and factual calibration. Delayed opportunities add four successes over immediate-only choices, and factual calibration adds three, on 228 paired episodes. Removing the temporal survival bound changes neither successes nor terminal durations in that screen. The complete counts and the earlier random-state failures remain in Appendix A; these controls establish the scope of the measured component effects rather than attributing a gain to every proposed safeguard.

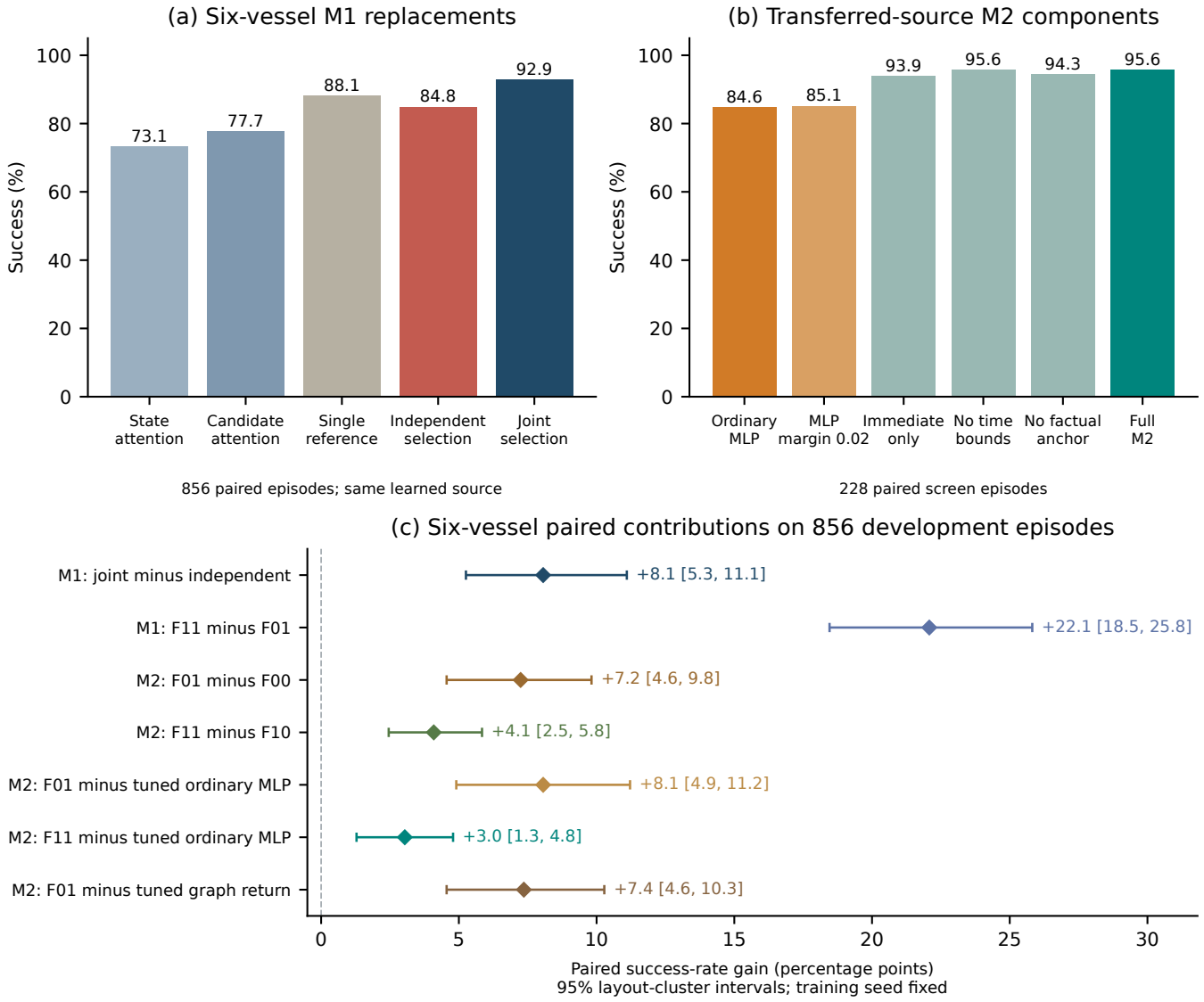


Fig. 6. Direct mechanism comparisons. (a) Six-vessel-source neural replacements, single-reference prediction, and independent/joint selectors on 856 paired episodes; the two neural alternatives share measured supervision, capacity, and update budget. (b) Transferred-source conditional-value component controls on 228 paired screen episodes. (c) Six-vessel-source contributions on 856 paired development episodes: matched joint versus independent selection, both four-configuration contrasts, each conditional model against its own tuned ordinary MLP, and F01 against the matched graph-return control. The graph-return contrast uses a common batched executor. Error bars are 95% layout-cluster bootstrap intervals.

D. Agility, fleet size, and opponent adaptation

The retained transferred-source scope experiments use the same methods and paired scene design across attacker agility and fleet size. Cross-fleet inference uses all current vessels in the shared neural potential, projects valid intervention roles to the current fleet, and generates every single-vessel alternative for that fleet size. Six-vessel empirical calibration is not transferred as if it were a measurement at another fleet size.

In the completed 16-scene screens with three, four, five, seven, and eight defenders, the complete method succeeds in 16, 15, 15, 12, and 10 episodes, respectively. The joint predecessor obtains 14, 15, 15, 14, and 10 successes. Conditional value therefore adds two successes at three defenders, preserves the counts at four, five, and eight, and loses two at seven. Refined joint selection obtains 14, 15, 15, 13, and 12 successes, compared with 14, 16, 16, 11, and 9 for independent selection. These paired screens identify where each mechanism transfers and where its current mapping loses performance.

Opponent adaptation uses one common learned attacker trained against a fixed source defender. All defender methods face the same frozen attacker before adaptation. New paired continuations are then collected under that opponent and mixed with identified APF training observations. The same adapted defenders are reevaluated against both the learned attacker and the original APF opponent, so recovery and retention remain separate outcomes.

On the 64 adaptation-development layouts, the complete method improves from 49 to 59 successes. This improvement does not transfer to 64 new paired layouts: the complete method obtains 54 successes before adaptation and 53 after-

ward, while the matched ordinary MLP obtains 52 in both conditions. Conditional value alone changes from 22 to 20 successes. On the original 856 APF episodes, the adapted complete method retains 805 successes, compared with 815 before adaptation; the ordinary MLP changes from 692 to 673, and conditional value alone from 296 to 281. Thus this adaptation recipe improves its fitting-layout performance but does not establish recovery on new layouts or full retention against the original opponent.

E. High-fidelity closed-loop comparison

The retained transferred-source VRX comparison [25] executes the current learned–Boids commands in two settings with agility 2.5 and 3.5 and two scene seeds per condition. Each method runs all eight scenes. VRX uses a 0.2-s control period, 15-m target radius, 5-m collision radius, 5.5-m capture radius, and 100-s time limit. These settings are reported separately from the numerical 80-s task.

All 80 completed trials passed the execution-validity checks. The complete reference method captures in 7/8 scenes and breaches in one; its matched single-intervention MLP captures in 7/8 and collides in one. Refined joint fusion captures in 7/8 and collides in one, the independent selector in 4/8, conditional value alone in 6/8, and the transferred source policy in 2/8. The factual joint predecessor captures in all eight scenes. An earlier ordinary-return comparator with a different predecessor and execution budget also captures in all eight; that result remains separate from the matched comparisons.

After additional paired short-response training, the refitted complete method captures in 8/8 scenes and its matched refitted MLP in 6/8. The two additional captures rescue one collision and one target breach without harming a matched MLP success. The refit also captures the scene where the reference complete method breached, as shown in Fig. 8. These eight development scenes establish a matched-comparator advantage for the refit, while its success count remains equal to the factual joint predecessor. Its mean capture time is 12.9 s versus 9.8 s for the predecessor, so this batch does not establish an additional efficiency gain from mechanism two. The valid reference failure remains in the comparison.

VII. Discussion

The two mechanisms solve related but distinct problems. Joint fusion coordinates physically compatible candidate controls; conditional value assesses whether a finite change improves the eventual task. A large gain over a transferred source policy alone cannot establish both contributions. The matched joint-selection and ordinary-return controls provide the more specific evidence.

The shared potential makes local differences consistent with complete-plan values, but consistency does not guarantee accurate generalization. Factual calibration can prevent avoidable damage near observed states, while timing distinguishes an executable rescue from a change that starts too late. New-state paired continuations and the internal component controls test these roles directly. Inference cost includes the neural potential, factual bank, and paired residual calibration; it is reported in addition to trainable parameter count.

The development evidence separates reliability, mechanism contribution, and ranking accuracy. Joint-selection gains vary across independently learned source policies. Conditional value has a clear random-candidate ranking advantage, while its deployed choices at uniformly sampled states add only one successful continuation. The transferred-source VRX refit matches its factual predecessor's success count while taking longer to capture. These outcomes motivate broader intervention coverage and stronger coordination across source policies. Final generalization claims use the frozen-method held-out comparison and complete independent training runs.

VIII. Conclusion

Joint fusion and conditional intervention value learning provide two complementary ways to improve cooperative target defense. The first selects compatible fleet mixtures from common learned and Boids candidates. The second learns whether and when a local change improves complete closed-loop outcomes, using paired differences tied to one task potential. Completed development comparisons show joint-selection gains of 10.7 points with the transferred source and 8.1 points with a fully trained six-vessel source. The complete six-vessel-source method achieves 97.0% success for training seed 42. Its conditional value adds 4.1 points over the joint predecessor and 3.0 over a tuned matched ordinary MLP, with a 24.2-point advantage in random-candidate ranking on new states. The remaining empirical scope is determined by independent full-pipeline training, frozen held-out evaluation, opponent adaptation, and the retained high-fidelity successes and failures.

Appendix A
Transferred-Source Mechanism Evidence

These completed comparisons use a source trained with three defenders and transferred to six. They remain distinct from the fully trained six-vessel-source comparison in the main text. Their paired rescue and harm counts, tuned ordinary-return controls, random-state failures, and learning variability are retained in full.

A. Complete method and four configurations

The transferred-source reference succeeds in 815 of 856 episodes (95.2%), compared with 738 (86.2%) for its frozen joint-fusion predecessor. It rescues 87 predecessor failures and harms 10 predecessor successes, adding 9.0 points [6.9,11.2]. Its F01 configuration also improves on F00, demonstrating useful interventions without joint fusion. This completed four-configuration experiment is retained separately from the new six-vessel-source pipeline.

Of the complete method's 815 successful defenses, 813 end in capture and two at the defense deadline. Compared with its factual predecessor, collisions decrease from 74 to 30 and source-defined intrusion failures from 44 to 11. The capture-only increment is 8.8 points [6.7,11.1]. On the 727 episodes captured by both controllers, the paired mean capture-time

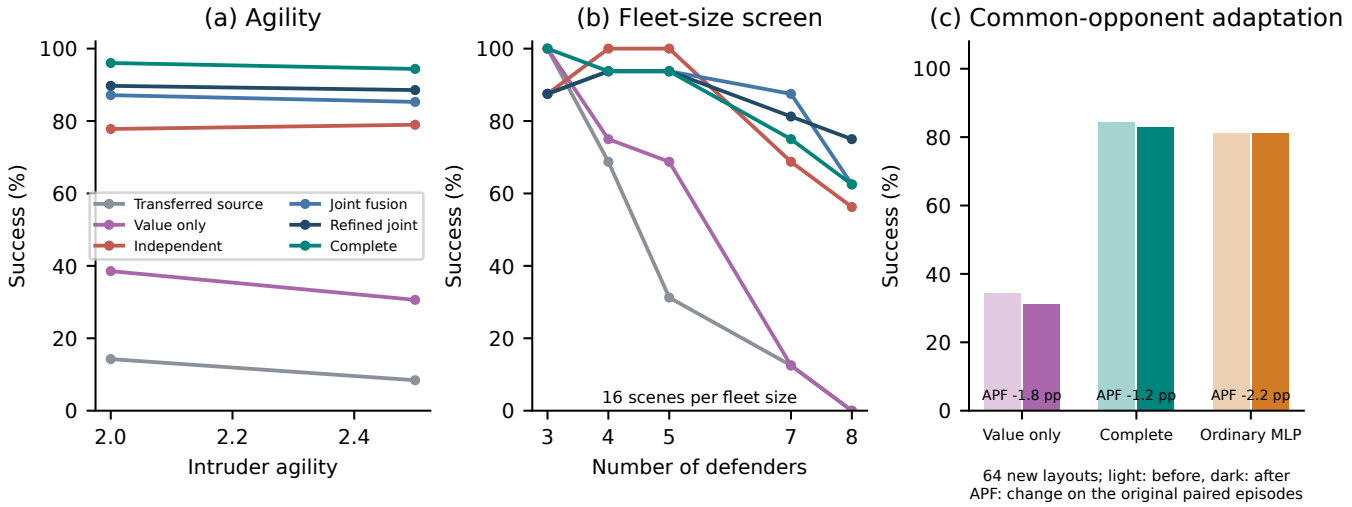


Fig. 7. Transferred-source scope of the two mechanisms: paired agility comparisons, explicit cross-fleet mappings on 16 scenes per fleet size, and common-opponent adaptation on 64 new paired layouts. The adaptation panel also reports the change in success rate on the original APF episodes.

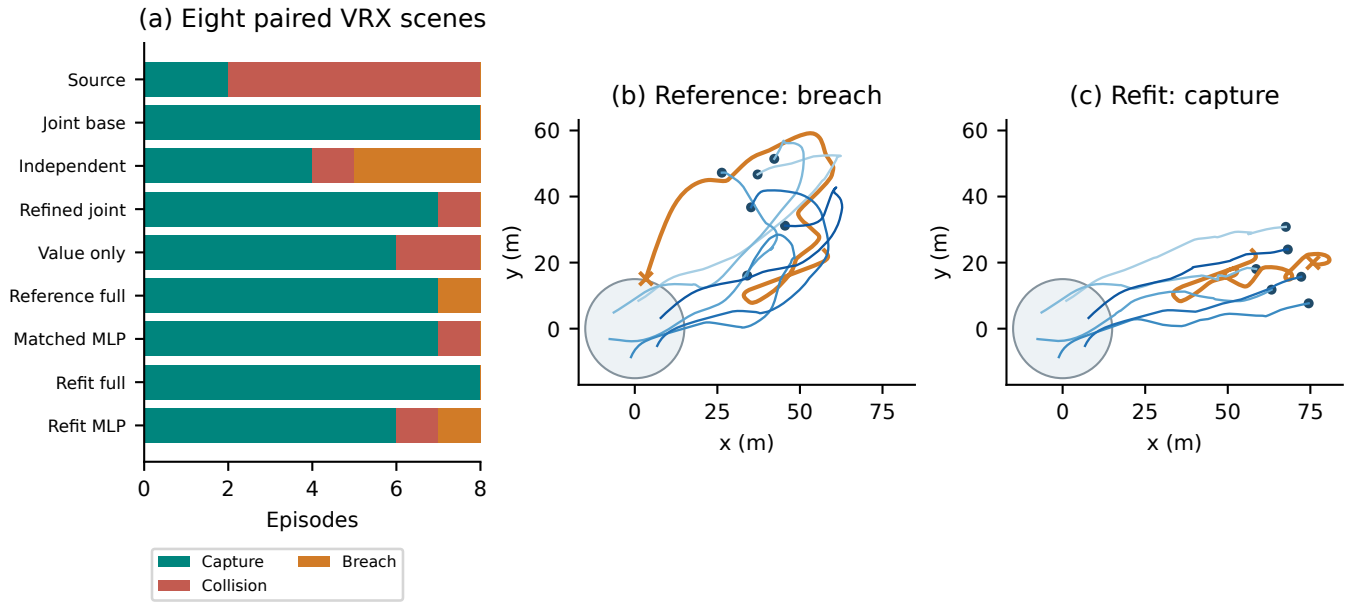


Fig. 8. Eight-scene paired transferred-source VRX development comparison. Reference and short-response refitted complete methods each have a matched ordinary-return comparator. The trajectories retain the reference method's valid breach and show the refitted method's capture on the same scene.

difference is -0.02 s $[-0.39,0.33]$; the demonstrated benefit is increased capture reliability and fewer failures.

The refined standalone joint selector obtains 763 successes (89.1%). Fresh complete continuations under this stronger predecessor train the conditional refit, which obtains 814 successes (95.1%). It rescues 65 predecessor failures and harms 14 predecessor successes, adding 6.0 points [4.0,8.2]. Collisions decrease from 63 to 25 and source-defined failures from 30 to 17. The paired capture-time difference on 748 jointly captured episodes is -0.001 s $[-0.26,0.24]$. Thus strengthening the predecessor preserves a separate conditional-value increment.

B. Joint selection and neural replacements

The refined joint selector obtains 92 more successes than the independent selector across the same 856 episodes, an absolute gain of 10.7 points (development layout-cluster interval [7.5,14.1]). It rescues 144 independent failures while harming 52 independent successes. Its per-suite increments are 12.1, 11.5, 9.5, and 9.5 points. Because both methods receive the same nine candidates per vessel and the same prediction tensor, this contrast directly measures the value of resolving their combination jointly.

Joint selection increases captures from 669 to 762, reduces defender collisions from 114 to 63, and reduces source-defined intrusion failures from 71 to 30. The collision-rate reduction is

TABLE III
RETAINED TRANSFERRED-SOURCE DEVELOPMENT COMPARISONS. ALL METHODS USE A THREE-VESSEL SOURCE DEPLOYED WITH SIX DEFENDERS. ORDINARY MLP ROWS BELOW USE THEIR ORIGINAL MAXIMUM-RETURN DECISION RULE; STRONGER THRESHOLDED COMPARISONS ARE REPORTED IN THE TEXT.

Method	Source /256	4310 /200	Earlier /200	Later /200	Total /856
F00: source	37	21	14	25	97
F10: joint fusion	225	171	169	173	738
F01: conditional value	111	57	64	64	296
F11: complete	254	175	196	190	815
Matched ordinary MLP	221	163	157	151	692
Independent fusion	205	153	153	160	671
Refined joint fusion	236	176	172	179	763
Refitted complete	245	187	193	189	814
Refitted ordinary MLP	231	167	162	165	725

6.0 points [3.3,8.8]. On the 616 jointly captured episodes, its paired mean capture time changes by -0.24 s [-1.27,0.77].

The original source backbone and four independently trained source backbones give five separate joint-versus-independent comparisons on the common 228-episode screen. Joint selection obtains 948/1,140 successes, compared with 843/1,140 for independent selection. The mean gain is 9.2 points; individual gains are 12.3, 4.4, 7.9, 10.1, and 11.4 points. All four additional backbones complete the original one-million-step source training. This experiment repeats the analytic joint selector across independently learned source commands; it does not yet constitute five retrainings of the complete value-learning pipeline.

The strengthened state-attention and candidate-attention alternatives obtain 446/856 (52.1%) and 551/856 (64.4%) successes. Both use two attention layers of width 128, 192,769 trainable parameters, 24,000 optimizer updates, and the same 24,576 measured teacher queries collected on teacher and student trajectories. Training balances changed and unchanged teacher choices and retains every physically equivalent candidate label; candidate trajectories are expressed in a common task frame. The joint selector supplies supervision only during training. At deployment, each replacement selects candidates through its own neural outputs. These fitted alternatives improve substantially over their initial versions but remain below independent selection (78.4%) and joint selection (89.1%). The matched joint-versus-independent contrast provides the direct evidence for joint decision making; the replacement comparison additionally tests whether these neural representations recover its benefit.

C. Conditional value, replacement learning, and component controls

The complete reference controller exceeds the matched ordinary MLP by 123 successes, or 14.4 points (development layout-cluster interval [11.6,17.4]). The paired outcomes contain 139 rescues and 16 harms. The per-suite gains are 12.9, 6.0, 19.5, and 19.5 points. The smaller 4310 increment is retained because it is informative about generalization beyond the fitting layouts.

The conditional model produces 813 captures and 30 collisions, compared with 692 captures and 114 collisions for the MLP. Its capture-only gain is 14.1 points [11.2,17.3], and its collision-rate reduction is 9.8 points [7.4,12.5]. Among the 674 episodes captured by both, the paired capture-time difference is 0.69 s [-0.14,1.49].

Giving that ordinary MLP a positive gain threshold improves its results without changing its learned weights. Thresholds of 0.02, 0.08, and 0.12 yield 729, 744, and 752 successes, respectively. Against the strongest of these completed controls, the reference method retains a 7.4-point gain [5.0,9.8], with 77 rescues and 14 harms. The larger 14.4-point contrast above therefore describes the maximum-return decision rule specifically, rather than the best thresholded ordinary controller.

The stronger-predecessor conditional refit obtains 814 successes against 725 for its freshly trained matched MLP, a gain of 10.4 points [7.9,13.0], with 106 rescues and 17 harms. Captures increase from 723 to 813 and collisions decrease from 98 to 25. On 705 jointly captured episodes, its capture-time difference is 0.17 s [-0.51,0.82]. Both models use the same newly collected labels and 30,000 updates. The ordinary model selected with a 0.08 gain margin subsequently reaches 771 successes, reducing this refit contrast to 5.0 points. Tuning conditional confidence on the same screen produces 807 successes, below the original 814; that screen-selected variant therefore does not replace the stronger completed conditional result.

Across five independent value fits, the conditional model succeeds in 1,075 of 1,140 paired evaluation episodes and the ordinary MLP in 959, a mean increment of 10.2 points. Each run contains the same 128-source/100-4310 screen, so these are repeated learning outcomes on a shared development collection. The per-run increments are 11.0, 7.9, 9.2, 13.6, and 9.2 points. Seed 44 reverses the advantage on the 4310 subset (85 versus 89 successes); its combined advantage does not erase that adverse result.

Adding paired short-response observations and 6,000 updates to both estimators gives 813/856 successes for conditional value and 684/856 for its matched MLP, a 15.1-point difference with 143 paired rescues and 14 harms. Both refits receive the same 82,857 terminal observations and 30,000 cumulative updates. This refit strengthens the matched-MLP contrast but remains two successes below the reference complete method; its separate VRX gains are reported below.

New-state value validation samples one eligible pre-action state uniformly from each of 64 complete factual trajectories on new layouts. At every root, the conditional model and ordinary MLP propose their own action; both choices and the factual action are evaluated to termination under eight new shared futures. The conditional model intervenes at 12 roots and yields 408 successful continuations, versus 400 for the MLP and 411 for the factual controller out of 512 continuations. Relative to the factual controller, conditional decisions rescue five futures and harm eight; the MLP rescues 14 and harms 25. Conditional value therefore reduces harmful changes, but neither learned selector improves on factual continuation in this random-state experiment. Positive predicted gains frequently correspond to zero measured gain, identifying

a remaining calibration and state-coverage weakness. Component controls separately remove delayed choices, temporal survival bounds, and factual calibration while retaining learned parameters and the common deployment budget.

A separate collection measures six outcome-independent random candidates at each of 64 new states under eight shared futures. Of these states, 54 have at least one pair with different empirical success probabilities, yielding 574 nontied comparisons. Conditional value agrees with the measured ordering in 58.9% of comparisons, versus 57.2% for the ordinary MLP; the paired difference is 1.7 points with a stratified state-bootstrap interval of $[-5.2,9.0]$ points. Its gain-prediction root mean squared error is 0.247, compared with 0.402 for the MLP. Thus the conditional estimator improves gain calibration in this collection, while a ranking advantage over ordinary return learning remains unresolved. All candidates and neutral outcomes remain in the calibration calculation; only outcome ties are excluded from the ordering score.

The stronger-predecessor refit is tested separately on 64 new layouts with eight shared futures and three outcome-independent random candidates per root. Its executed choices yield 422/512 successful continuations, compared with 417 for factual continuation and 413 for the MLP. Conditional choices rescue six factual failures and harm one factual success; ordinary choices rescue 13 and harm 17. Among 121 nontied random-candidate comparisons from 48 informative roots, the refit's ordering concordance is 66.9% [57.7,75.8], versus 54.1% [44.8,63.4] for the MLP. Their paired difference is 12.8 points [1.7,23.3]. Gain mean squared error is 0.047 versus 0.155. This new-state experiment links improved ranking and fewer harmful changes to a positive realized intervention gain under the stronger predecessor.

On the 228-episode component screen, the complete model achieves 218 successes, immediate-only choices 214, removal of temporal bounds 218, and removal of factual calibration 215. Relative to immediate-only choices, the complete model rescues eight episodes and harms four. Factual calibration rescues three and harms none. Removing temporal bounds changes neither success nor terminal duration in this screen; these completed outcomes therefore provide no separate performance gain for that bound. All three controls achieve the same 91/100 successes on the 4310 subset, locating the observed component gains in the source-development subset.

REFERENCES

[1] Y. Qiao et al., "Survey of deep learning for autonomous surface vehicles in marine environments," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 3678–3701, 2023, doi: 10.1109/TITS.2023.3235911.

[2] H. Fu and H. H.-T. Liu, "Guarding a territory against an intelligent intruder: Strategy design and experimental verification," *IEEE/ASME Trans. Mechatron.*, vol. 25, no. 4, pp. 1765–1772, 2020, doi: 10.1109/TMECH.2020.2996901.

[3] Y. Chen et al., "A distributed pipeline for collaborative pursuit in the target guarding problem," *IEEE Robot. Autom. Lett.*, vol. 9, no. 3, pp. 2064–2071, 2024, doi: 10.1109/LRA.2024.3349977.

[4] R. Yan et al., "Multiplayer homicidal chauffeur reach-avoid games: A pursuit enclosure function approach," *Automatica*, vol. 167, Art. no. 111770, 2024, doi: 10.1016/j.automatica.2024.111770.

[5] X. Fang, C. Wang, L. Xie, and J. Chen, "Cooperative pursuit with multi-pursuer and one faster free-moving evader," *IEEE Trans. Cybern.*, vol. 52, no. 3, pp. 1405–1414, 2022, doi: 10.1109/TCYB.2019.2958548.

[6] X. Guo, R. Cui, and W. Yan, "Pursuit-evasion games of marine surface vessels using neural network-based control," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 55, no. 1, pp. 18–27, 2025, doi: 10.1109/TSMC.2023.3347044.

[7] M. Zhou, Z. Wang, J. Wang, and Z. Cao, "Multi-robot collaborative hunting in cluttered environments with obstacle-avoiding Voronoi cells," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 7, pp. 1643–1655, 2024, doi: 10.1109/JAS.2023.124041.

[8] C. de Souza et al., "Decentralized multi-agent pursuit using deep reinforcement learning," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 4552–4559, 2021, doi: 10.1109/LRA.2021.3068952.

[9] R. Zhang et al., "Game of drones: Multi-UAV pursuit-evasion game with online motion planning by deep reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7900–7909, 2023, doi: 10.1109/TNNLS.2022.3146976.

[10] S. Xue et al., "Multi-agent self-attention reinforcement learning for multi-USV hunting target," *Neural Netw.*, vol. 189, Art. no. 107574, 2025, doi: 10.1016/j.neunet.2025.107574.

[11] X. Lin, P. Szenher, Y. Huang, and B. Englot, "Distributional reinforcement learning based integrated decision making and control for autonomous surface vehicles," *IEEE Robot. Autom. Lett.*, vol. 10, no. 2, pp. 1194–1201, 2025, doi: 10.1109/LRA.2024.3518839.

[12] M. Hüttenrauch, A. Šošić, and G. Neumann, "Deep reinforcement learning for swarm systems," *J. Mach. Learn. Res.*, vol. 20, no. 54, pp. 1–31, 2019. https://jmlr.org/papers/v20/18-476.html

[13] T. Rashid et al., "Monotonic value function factorisation for deep multi-agent reinforcement learning," *J. Mach. Learn. Res.*, vol. 21, no. 178, pp. 1–51, 2020. https://jmlr.org/papers/v21/20-081.html

[14] Z. Ma, Y. Luo, and J. Pan, "Learning selective communication for multi-agent path finding," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 1455–1462, 2022, doi: 10.1109/LRA.2021.3139145.

[15] Q. Li, W. Lin, Z. Liu, and A. Prorok, "Message-aware graph attention networks for large-scale multi-robot path planning," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 5533–5540, 2021, doi: 10.1109/LRA.2021.3077863.

[16] K. Rana et al., "Bayesian controller fusion: Leveraging control priors in deep reinforcement learning for robotics," *Int. J. Robot. Res.*, vol. 42, no. 3, pp. 123–146, 2023, doi: 10.1177/02783649231167210.

[17] S. H. Jeon, H. J. Lee, S. Hong, and S. Kim, "Residual MPC: Blending reinforcement learning with GPU-parallelized model predictive control," *IEEE Trans. Robot.*, vol. 42, pp. 3469–3488, 2026, doi: 10.1109/TRO.2026.3721669.

[18] J. Tao, T. Shen, D. Zhao, and F. Zhang, "ARBoids: Adaptive residual reinforcement learning with Boids model for cooperative multi-USV target defense," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 3637–3644, 2026, doi: 10.1109/LRA.2026.3662620.

[19] S. Bhattacharya et al., "Multiagent reinforcement learning: Rollout and policy iteration for POMDP with application to multirobot problems," *IEEE Trans. Robot.*, vol. 40, pp. 2003–2023, 2024, doi: 10.1109/TRO.2023.3347128.

[20] L. Brunke et al., "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annu. Rev. Control Robot. Auton. Syst.*, vol. 5, no. 1, pp. 411–444, 2022, doi: 10.1146/annurev-control-042920-020211.

[21] J. Yu et al., "A cooperative hunting method for multi-USVs based on trajectory prediction by OR-LSTM," *IEEE Trans. Veh. Technol.*, vol. 73, no. 12, pp. 18087–18101, 2024, doi: 10.1109/TVT.2024.3432739.

[22] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. ICML*, PMLR 80, pp. 1861–1870, 2018. https://proceedings.mlr.press/v80/haarnoja18b.html

[23] J. Foerster et al., "Counterfactual multi-agent policy gradients," in *Proc. AAAI*, 2018. https://arxiv.org/abs/1705.08926

[24] V. Feinberg et al., "Model-based value estimation for efficient model-free reinforcement learning," 2018. https://arxiv.org/abs/1803.00101

[25] Open Source Robotics Foundation, "Virtual RobotX (VRX) simulation environment," software repository. https://github.com/osrf/vrx

[26] T. Johannink et al., "Residual reinforcement learning for robot control," arXiv:1812.03201, 2018. https://arxiv.org/abs/1812.03201